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JEE MAIN · QUESTION PAPER

JEE Main 2022 (Online) 28th June Morning Shift

JEE Main Question Paper with Answer Key

28 Jun 2022	90	21	English
HELD ON	QUESTIONS	OUT OF SYLLABUS	LANGUAGE

INSIDE THIS PDF

Chemistry	23 questions
Mathematics	19 questions
Physics	27 questions
Out of syllabus	21 questions
Answer key	at the end

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Chemistry 23 questions

Q1. MCQ (Single Correct Answer)

Element "E" belongs to the period 4 and group 16 of the periodic table. The valence shell electron configuration of the element, which is just above "E" in the group is

(A) $3s^2, 3p^4$ (B) $3d^{10}, 4s^2, 4p^4$ (C) $4d^{10}, 5s^2, 5p^4$ (D) $2s^2, 2p^4$

Open solution

Q2. MCQ (Single Correct Answer)

Nitrogen gas is obtained by thermal decomposition of :

(A) $Ba(NO_3)_2$ (B) $Ba(N_3)_2$ (C) $NaNO_2$ (D) $NaNO_3$

Open solution

Q3. MCQ (Single Correct Answer)

Which one of the lanthanoids given below is the most stable in divalent form?

(A) Ce (Atomic Number 58) (B) Sm (Atomic number 62)

(C) Eu (Atomic Number 63) (D) Yb (Atomic Number 70)

Open solution

Q4. MCQ (Single Correct Answer)

Given below are two statements :

Statement I : $[Ni(CN)_4]^{2-}$ is square planar and diamagnetic complex, with dsp^2 hybridization for Ni but $[Ni(CO)_4]$ is tetrahedral, paramagnetic and with sp^3-hybridization for Ni.

Statement II : $[NiCl_4]^{2-}$ and $[Ni(CO)_4]$ both have same d-electron configuration have same geometry and are paramagnetic.

In light of the above statements, choose the correct answer from the options given below :

(A) Both Statement I and Statement II are true.

(B) Both Statement I and Statement II are false.

(C) Statement I is correct but Statement II is false.

(D) Statement I is false but Statement II is correct.

Open solution

Q5. MCQ (Single Correct Answer)

Which one of the following techniques is not used to spot components of a mixture separated on thin layer chromatographic plate?

(A) I_2 (Solid)

(B) U.V. Light

(C) Visualisation agent as a component of mobile phase

(D) Spraying of an appropriate reagent

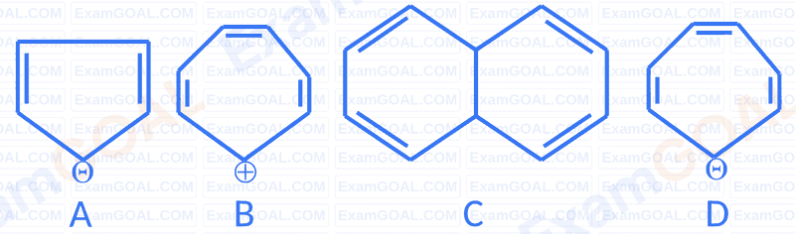
Open solution

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Q6. MCQ (Single Correct Answer)

Open solution

Which of the following structures are aromatic in nature?



(A) A, B, C, and D **(B)** Only A and B **(C)** Only A and C **(D)** Only B, C and D

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Q7. MCQ (Single Correct Answer)

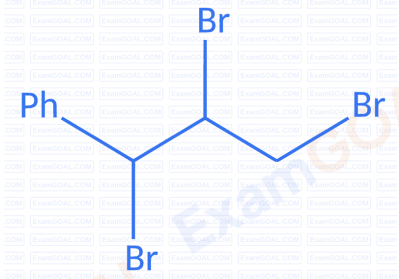
Open solution

The major product (P) in the reaction

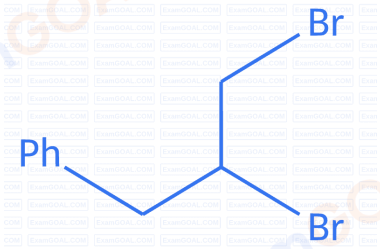


is

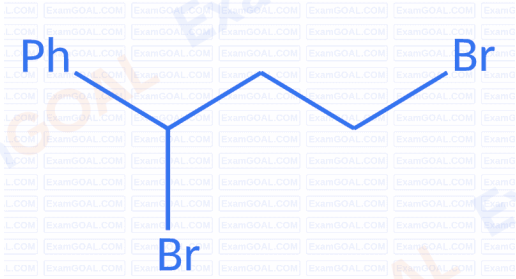
(A)



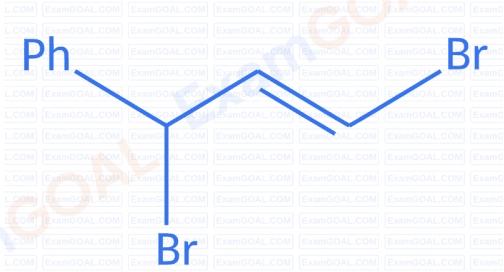
(B)



(C)



(D)



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Q8. MCQ (Single Correct Answer)

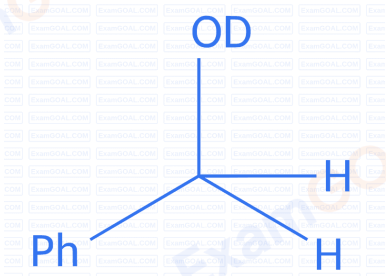
Open solution

The correct structure of product 'A' formed in the following reaction.

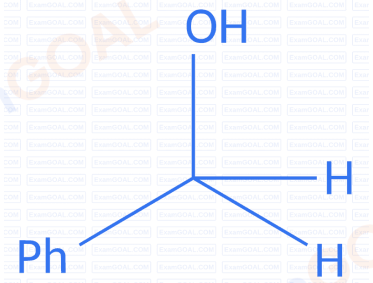
$$PhCHO + Ph\,.\,CHO \xrightarrow[\text{in } D_2O]{NaOD} A + Ph - \overset{\overset{O}{\|}}{C} - O^- \ (Ph \text{ is } -C_6H_5)$$

is

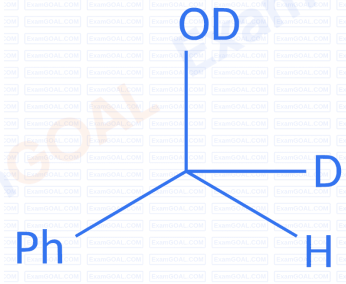
(A)



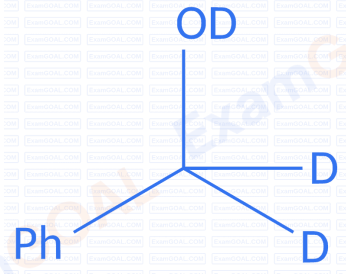
(B)



(C)



(D)



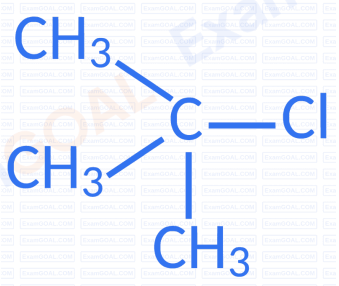
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Q9. MCQ (Single Correct Answer)

Open solution

Which one of the following compounds is inactive towards S_N1 reaction?

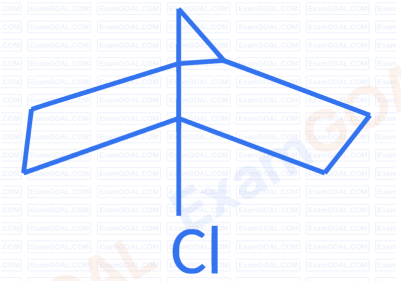
(A)



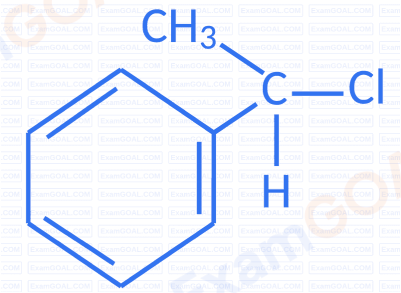
(B)



(C)



(D)

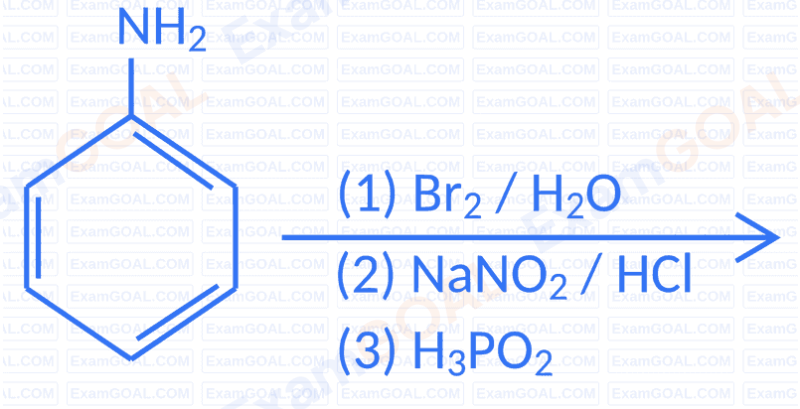


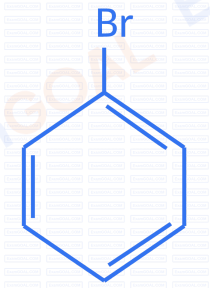
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Q10. MCQ (Single Correct Answer)

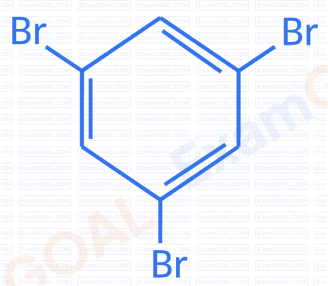
Open solution

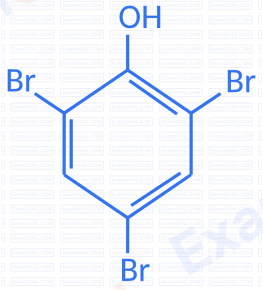
Identify the major product formed in the following sequence of reactions:



(A) 

(B) 

(C) 

(D) 

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Q11. MCQ (Single Correct Answer)

A primary aliphatic amine on reaction with nitrous acid in cold (273 K) and there after raising temperature of reaction mixture to room temperature (298 K), gives a/an

(A) nitrile (B) alcohol (C) diazonium salt (D) secondary amine

Open solution

Q12. MCQ (Single Correct Answer)

Stability of α-Helix structure of proteins depends upon

(A) dipolar interaction
(B) H-bonding interaction
(C) van der Waals forces
(D) π-stacking interaction

Open solution

Q13. MCQ (Single Correct Answer)

The formula of the purple colour formed in Lassaigne's test for sulphur using sodium nitroprusside is :

(A) $NaFe[Fe(CN)_6]$
(B) $Na[Cr(NH_3)_2(NCS)_4]$
(C) $Na_2[Fe(CN)_5(NO)]$
(D) $Na_4[Fe(CN)_5(NOS)]$

Open solution

Q14. Numerical

A 2.0 g sample containing MnO_2 is treated with HCl liberating Cl_2. The Cl_2 gas is passed into a solution of KI and 60.0 mL of 0.1 M $Na_2S_2O_3$ is required to titrate the liberated iodine. The percentage of MnO_2 in the sample is _____. (Nearest integer)

[Atomic masses (in u) Mn = 55; Cl = 35.5; O = 16, I = 127, Na = 23, K = 39, S = 32]

Open solution

Q15. Numerical

If the work function of a metal is 6.63×10^{-19}J, the maximum wavelength of the photon required to remove a photoelectron from the metal is _____ nm. (Nearest integer)

[Given : h = 6.63×10^{-34} J s, and c = 3×10^{8} m s^{-1}]

Open solution

Q16. Numerical

The hybridization of P exhibited in PF_5 is sp^xd^y. The value of y is _____.

Open solution

Q17. Numerical

4.0 L of an ideal gas is allowed to expand isothermally into vacuum until the total volume is 20 L. The amount of heat absorbed in this expansion is _____ L atm.

Open solution

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Q18. Numerical

The vapour pressures of two volatile liquids A and B at 25°C are 50 Torr and 100 Torr, respectively. If the liquid mixture contains 0.3 mole fraction of A, then the mole fraction of liquid B in the vapour phase is $\frac{x}{17}$. The value of x is _____.

Open solution

Q19. Numerical

The solubility product of a sparingly soluble salt A_2X_3 is 1.1×10^{-23}. If specific conductance of the solution is 3×10^{-5} S m^{-1}, the limiting molar conductivity of the solution is $x \times 10^{-3}$ S m^2 mol^{-1}. The value of x is _____.

Open solution

Q20. Numerical

The quantity of electricity in Faraday needed to reduce 1 mol of $Cr_2O_7^{2-}$ to Cr^{3+} is _____.

Open solution

Q21. Numerical

For a first order reaction A $\rightarrow$ B, the rate constant, k = 5.5×10^{-14} s^{-1}. The time required for 67% completion of reaction is x $\times 10^{-1}$ times the half life of reaction. The value of x is _____ (Nearest integer)

(Given : log 3 = 0.4771)

Open solution

Q22. Numerical

Number of complexes which will exhibit synergic bonding amongst, $[Cr(CO)_6]$, $[Mn(CO)_5]$ and $[Mn_2(CO)_{10}]$ is _____.

Open solution

Q23. Numerical

In the estimation of bromine, 0.5 g of an organic compound gave 0.40 g of silver bromide. The percentage of bromine in the given compound is _____ % (nearest integer)

(Relative atomic masses of Ag and Br are 108u and 80u, respectively).

Open solution

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Mathematics 19 questions

Q24. MCQ (Single Correct Answer)

Let a function f : N $\rightarrow$ N be defined by

$$f(n)=\begin{cases}2n, & n=2,4,6,8,\ldots\ldots\\ n-1, & n=3,7,11,15,\ldots\ldots\\ \frac{n+1}{2}, & n=1,5,9,13,\ldots\ldots\end{cases}$$

then, f is

(A) one-one but not onto

(B) onto but not one-one

(C) neither one-one nor onto

(D) one-one and onto

Open solution

Q25. MCQ (Single Correct Answer)

If the system of linear equations

$2x+3y-z=-2$

$x+y+z=4$

$x-y+|\lambda|z=4\lambda-4$

where, $\lambda \in$ R, has no solution, then

(A) λ = 7

(B) λ = -7

(C) λ = 8

(D) λ^2 = 1

Open solution

Q26. MCQ (Single Correct Answer)

The total number of 5-digit numbers, formed by using the digits 1, 2, 3, 5, 6, 7 without repetition, which are multiple of 6, is :

(A) 36

(B) 48

(C) 60

(D) 72

Open solution

Q27. MCQ (Single Correct Answer)

Let $A_1, A_2, A_3, \ldots\ldots$ be an increasing geometric progression of positive real numbers. If $A_1A_3A_5A_7 = \frac{1}{1296}$ and $A_2 + A_4 = \frac{7}{36}$, then the value of $A_6 + A_8 + A_{10}$ is equal to

(A) 33

(B) 37

(C) 43

(D) 47

Open solution

Q28. MCQ (Single Correct Answer)

Let [t] denote the greatest integer less than or equal to t. Then, the value of the integral $\int_0^1 [-8x^2+6x-1]dx$ is equal to :

(A) -1

(B) $\frac{-5}{4}$

(C) $\frac{\sqrt{17}-13}{8}$

(D) $\frac{\sqrt{17}-16}{8}$

Open solution

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Q29. MCQ (Single Correct Answer)

Open solution

Let f : R $\rightarrow$ R be defined as

$$f(x)=\begin{cases}[e^x], & x<0\\ ae^x+[x-1], & 0\le x<1\\ b+[\sin(\pi x)], & 1\le x<2\\ [e^{-x}]-c, & x\ge 2\end{cases}$$

where a, b, c $\in$ R and [t] denotes greatest integer less than or equal to t. Then, which of the following statements is true?

(A) There exists a, b, c $\in$ R such that f is continuous on R.

(B) If f is discontinuous at exactly one point, then a + b + c = 1

(C) If f is discontinuous at exactly one point, then a + b + c $\neq$ 1

(D) f is discontinuous at at least two points, for any values of a, b and c

Q30. MCQ (Single Correct Answer)

Open solution

The area of the region S = {(x, y) : $y^2 \le 8x$, $y \ge \sqrt{2}x$, $x \ge 1$} is

(A) $\frac{13\sqrt{2}}{6}$ (B) $\frac{11\sqrt{2}}{6}$ (C) $\frac{5\sqrt{2}}{6}$ (D) $\frac{19\sqrt{2}}{6}$

Q31. MCQ (Single Correct Answer)

Open solution

Let the solution curve $y=y(x)$ of the differential equation

$$\left[\frac{x}{\sqrt{x^2-y^2}}+e^{\frac{y}{x}}\right]x\frac{dy}{dx}=x+\left[\frac{x}{\sqrt{x^2-y^2}}+e^{\frac{y}{x}}\right]y$$

pass through the points (1, 0) and (2α, α), α > 0. Then α is equal to

(A) $\frac{1}{2}\exp\left(\frac{\pi}{6}+\sqrt{e}-1\right)$ (B) $\frac{1}{2}\exp\left(\frac{\pi}{6}+e-1\right)$

(C) $\exp\left(\frac{\pi}{6}+\sqrt{e}+1\right)$ (D) $2\exp\left(\frac{\pi}{3}+\sqrt{e}-1\right)$

Q32. MCQ (Single Correct Answer)

Open solution

Let y = y(x) be the solution of the differential equation $x(1-x^2)\frac{dy}{dx}+(3x^2y-y-4x^3)=0, x>1$, with $y(2)=-2$. Then y(3) is equal to :

(A) -18 (B) -12 (C) -6 (D) -3

Q33. MCQ (Single Correct Answer)

Open solution

The number of real solutions of

$x^7+5x^3+3x+1=0$ is equal to _____.

(A) 0 (B) 1 (C) 3 (D) 5

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Q34. MCQ (Single Correct Answer)

The probability, that in a randomly selected 3-digit number at least two digits are odd, is :

(A) $\frac{19}{36}$ (B) $\frac{15}{36}$ (C) $\frac{13}{36}$ (D) $\frac{23}{36}$

Open solution

Q35. Numerical

Let R_1 and R_2 be relations on the set {1, 2, ..., 50} such that

$R_1 = \{(p, p^n) : p \text{ is a prime and } n \geq 0 \text{ is an integer}\}$ and

$R_2 = \{(p, p^n) : p \text{ is a prime and } n = 0 \text{ or } 1\}$.

Then, the number of elements in $R_1 - R_2$ is _____.

Open solution

Q36. Numerical

The number of real solutions of the equation $e^{4x} + 4e^{3x} - 58e^{2x} + 4e^x + 1 = 0$ is _____.

Open solution

Q37. Numerical

The mean and standard deviation of 15 observations are found to be 8 and 3 respectively. On rechecking it was found that, in the observations, 20 was misread as 5. Then, the correct variance is equal to _____.

Open solution

Q38. Numerical

If $\vec{a} = 2\hat{i} + \hat{j} + 3\hat{k}, \vec{b} = 3\hat{i} + 3\hat{j} + \hat{k}$ and $\vec{c} = c_1\hat{i} + c_2\hat{j} + c_3\hat{k}$ are coplanar vectors and $\vec{a} \cdot \vec{c} = 5, \vec{b} \perp \vec{c}$, then $122(c_1 + c_2 + c_3)$ is equal to _____.

Open solution

Q39. Numerical

A ray of light passing through the point P(2, 3) reflects on the x-axis at point A and the reflected ray passes through the point Q(5, 4). Let R be the point that divides the line segment AQ internally into the ratio 2 : 1. Let the co-ordinates of the foot of the perpendicular M from R on the bisector of the angle PAQ be (α, β). Then, the value of $7\alpha + 3\beta$ is equal to _____.

Open solution

Q40. Numerical

Let A = {1, a_1, a_2 ..... a_{18}, 77} be a set of integers with $1 < a_1 < a_2 < ..... < a_{18} < 77$.

Let the set A + A = {x + y : x, y $\in$ A} contain exactly 39 elements. Then, the value of $a_1 + a_2 + ... + a_{18}$ is equal to _____.

Open solution

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Q41. Numerical

The number of positive integers k such that the constant term in the binomial expansion of $\left(2x^3 + \frac{3}{x^k}\right)^{12}$, $x \neq 0$ is $2^8 \cdot l$, where l is an odd integer, is _____.

Open solution

Q42. Numerical

The number of elements in the set {z = a + ib $\in$ C : a, b $\in$ Z and 1 < | z − 3 + 2i | < 4} is _____.

Open solution

Physics 27 questions

Q43. MCQ (Single Correct Answer)

Given below are two statements : One is labelled as Assertion A and the other is labelled as Reason R.

Assertion A : Product of Pressure (P) and time (t) has the same dimension as that of coefficient of viscosity.

Reason R : Coefficient of viscosity = $\frac{Force}{Velocity\ gradient}$

Choose the correct answer from the options given below :

(A) Both A and R are true, and R is the correct explanation of A.

(B) Both A and R are true but R is NOT the correct explanation of A.

(C) A is true but R is false.

(D) A is false but R is true.

Open solution

Q44. MCQ (Single Correct Answer)

A particle of mass m is moving in a circular path of constant radius r such that its centripetal acceleration (a) is varying with time t as $a = k^2rt^2$, where k is a constant. The power delivered to the particle by the force acting on it is given as

(A) zero (B) $mk^2r^2t^2$ (C) mk^2r^2t (D) mk^2rt

Open solution

Q45. MCQ (Single Correct Answer)

Motion of a particle in x-y plane is described by a set of following equations $x = 4\sin\left(\frac{\pi}{2} - \omega t\right) m$ and $y = 4\sin(\omega t)\, m$. The path of the particle will be :

(A) circular (B) helical (C) parabolic (D) elliptical

Open solution

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Q46. MCQ (Single Correct Answer)

Open solution

Match List-I with List-II

	List-I		List-II
(A)	Moment of inertia of solid sphere of radius R about any tangent.	(I)	$\frac{5}{3}MR^2$
(B)	Moment of inertia of hollow sphere of radius (R) about any tangent.	(II)	$\frac{7}{5}MR^2$
(C)	Moment of inertia of circular ring of radius (R) about its diameter.	(III)	$\frac{1}{4}MR^2$
(D)	Moment of inertia of circular disc of radius (R) about any diameter.	(IV)	$\frac{1}{2}MR^2$

Choose the correct answer from the options given below :

(A) A - II, B - I, C - IV, D - III

(B) A - I, B - II, C - IV, D - III

(C) A - II, B - I, C - III, D - IV

(D) A - I, B - II, C - III, D - IV

Q47. MCQ (Single Correct Answer)

Open solution

Two planets A and B of equal mass are having their period of revolutions T_A and T_B such that $T_A = 2T_B$. These planets are revolving in the circular orbits of radii r_A and r_B respectively. Which out of the following would be the correct relationship of their orbits?

(A) $2r_A^2 = r_B^3$

(B) $r_A^3 = 2r_B^3$

(C) $r_A^3 = 4r_B^3$

(D) $T_A^2 - T_B^2 = \frac{\pi^2}{GM}\left(r_B^3 - 4r_A^3\right)$

Q48. MCQ (Single Correct Answer)

Open solution

A water drop of diameter 2 cm is broken into 64 equal droplets. The surface tension of water is 0.075 N/m. In this process the gain in surface energy will be :

(A) 2.8×10^{-4} J

(B) 1.5×10^{-3} J

(C) 1.9×10^{-4} J

(D) 9.4×10^{-5} J

Q49. MCQ (Single Correct Answer)

Open solution

Given below are two statements :

Statement I : When μ amount of an ideal gas undergoes adiabatic change from state (P_1, V_1, T_1) to state (P_2, V_2, T_2), then work done is $W = \frac{\mu R(T_2 - T_1)}{1-\gamma}$, where $\gamma = \frac{C_p}{C_v}$ and R = universal gas constant.

Statement II : In the above case, when work is done on the gas, the temperature of the gas would rise.

Choose the correct answer from the options given below :

(A) Both Statement I and Statement II are true.

(B) Both Statement I and Statement II are false.

(C) Statement I is true but Statement II is false.

(D) Statement I is false but Statement II is true.

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Q50. MCQ (Single Correct Answer)

Open solution

Given below are two statements :

Statement I : A point charge is brought in an electric field. The value of electric field at a point near to the charge may increase if the charge is positive.

Statement II : An electric dipole is placed in a non-uniform electric field. The net electric force on the dipole will not be zero.

Choose the correct answer from the options given below :

(A) Both Statement I and Statement II are true.

(B) Both Statement I and Statement II are false.

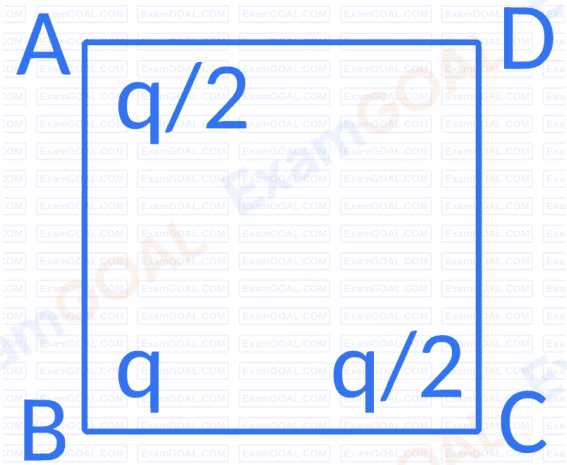
(C) Statement I is true but Statement II is false.

(D) Statement I is false but Statement II is true.

Q51. MCQ (Single Correct Answer)

Open solution

The three charges q/2, q and q/2 are placed at the corners A, B and C of a square of side 'a' as shown in figure. The magnitude of electric field (E) at the corner D of the square, is :



(A) $\frac{q}{4\pi\in_0 a^2}\left(\frac{1}{\sqrt{2}}+\frac{1}{2}\right)$

(B) $\frac{q}{4\pi\in_0 a^2}\left(1+\frac{1}{\sqrt{2}}\right)$

(C) $\frac{q}{4\pi\in_0 a^2}\left(1-\frac{1}{\sqrt{2}}\right)$

(D) $\frac{q}{4\pi\in_0 a^2}\left(\frac{1}{\sqrt{2}}-\frac{1}{2}\right)$

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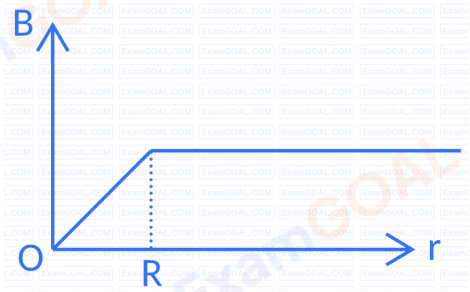
Q52. MCQ (Single Correct Answer)

Open solution

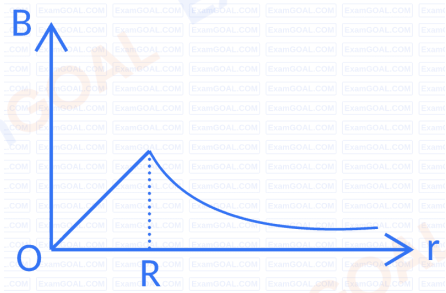
An infinitely long hollow conducting cylinder with radius R carries a uniform current along its surface.

Choose the correct representation of magnetic field (B) as a function of radial distance (r) from the axis of cylinder.

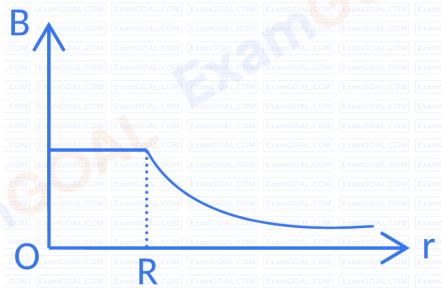
(A)



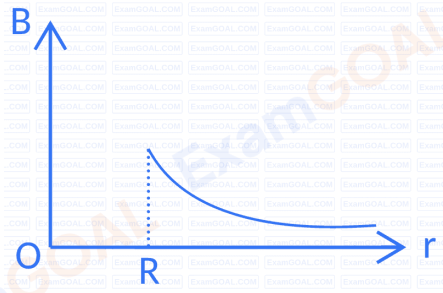
(B)



(C)



(D)



Q53. MCQ (Single Correct Answer)

Open solution

A radar sends an electromagnetic signal of electric field (E_0) = 2.25 V/m and magnetic field (B_0) = 1.5×10^{-8} T which strikes a target on line of sight at a distance of 3 km in a medium. After that, a part of signal (echo) reflects back towards the radar with same velocity and by same path. If the signal was transmitted at time t = 0 from radar, then after how much time echo will reach to the radar?

(A) 2.0×10^{-5} s (B) 4.0×10^{-5} s (C) 1.0×10^{-5} s (D) 8.0×10^{-5} s

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Q54. MCQ (Single Correct Answer)

The refracting angle of a prism is A and refractive index of the material of the prism is cot (A/2). Then the angle of minimum deviation will be -

(A) $180 - 2A$ (B) $90 - A$ (C) $180 + 2A$ (D) $180 - 3A$

Open solution

Q55. MCQ (Single Correct Answer)

The aperture of the objective is 24.4 cm. The resolving power of this telescope, if a light of wavelength 2440 $\overset{o}{A}$ is used to see the object will be :

(A) 8.1×10^6 (B) 10.0×10^7 (C) 8.2×10^5 (D) 1.0×10^{-8}

Open solution

Q56. MCQ (Single Correct Answer)

The de Broglie wavelengths for an electron and a photon are λ_e and λ_p respectively. For the same kinetic energy of electron and photon, which of the following presents the correct relation between the de Broglie wavelengths of two ?

(A) $\lambda_p \propto \lambda_e^2$ (B) $\lambda_p \propto \lambda_e$

(C) $\lambda_p \propto \sqrt{\lambda_e}$ (D) $\lambda_p \propto \sqrt{\frac{1}{\lambda_e}}$

Open solution

Q57. MCQ (Single Correct Answer)

The Q-value of a nuclear reaction and kinetic energy of the projectile particle, K_p are related as :

(A) $Q = K_p$ (B) $(K_p + Q) < 0$ (C) $Q < K_p$ (D) $(K_p + Q) > 0$

Open solution

Q58. MCQ (Single Correct Answer)

For using a multimeter to identify diode from electrical components, choose the correct statement out of the following about the diode :

(A) It is two terminal device which conducts current in both directions.

(B) It is two terminal device which conducts current in one direction only.

(C) It does not conduct current gives an initial deflection which decays to zero.

(D) It is three terminal device which conducts current in one direction only between central terminal and either of the remaining two terminals

Open solution

Q59. MCQ (Single Correct Answer)

The velocity of sound in a gas, in which two wavelengths 4.08 m and 4.16 m produce 40 beats in 12s, will be :

(A) 282.8 ms^{-1} (B) 175.5 ms^{-1} (C) 353.6 ms^{-1} (D) 707.2 ms^{-1}

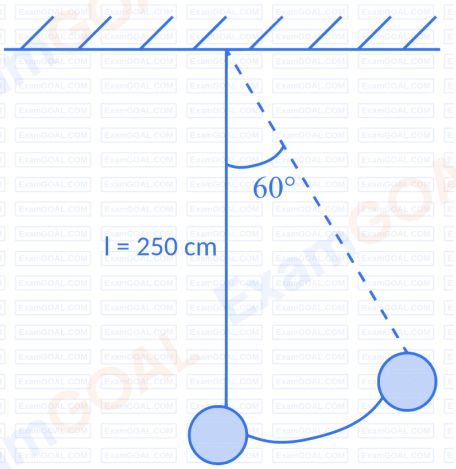
Open solution

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Q60. Numerical

Open solution

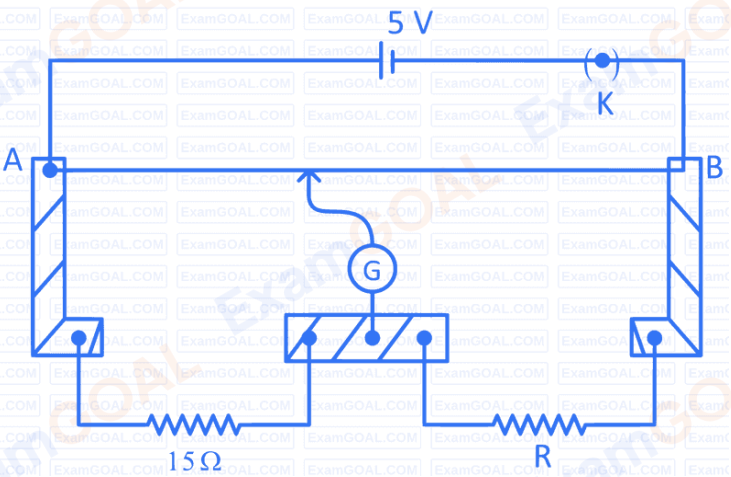
A pendulum is suspended by a string of length 250 cm. The mass of the bob of the pendulum is 200 g. The bob is pulled aside until the string is at 60° with vertical as shown in the figure. After releasing the bob, the maximum velocity attained by the bob will be _____ ms^{-1}. (if g = 10 m/s^2)



Q61. Numerical

Open solution

A meter bridge setup is shown in the figure. It is used to determine an unknown resistance R using a given resistor of 15 Ω. The galvanometer (G) shows null deflection when tapping key is at 43 cm mark from end A. If the end correction for end A is 2 cm, then the determined value of R will be _____ Ω.

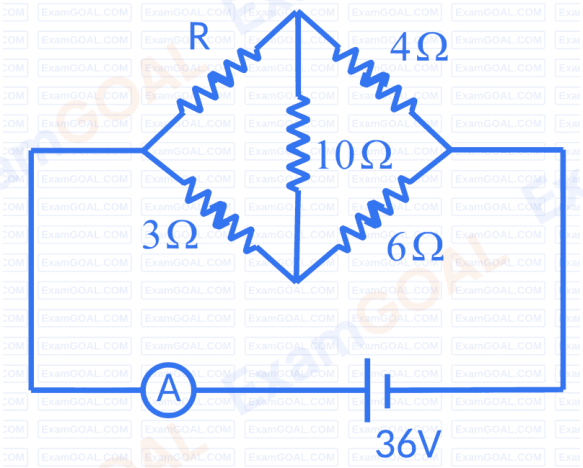


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Q62. Numerical

Current measured by the ammeter (A) in the reported circuit when no current flows through 10 Ω resistance, will be _____ A.

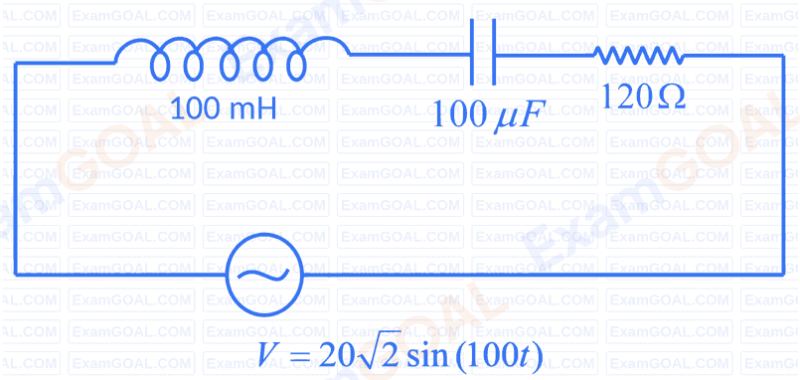
Open solution



Q63. Numerical

An AC source is connected to an inductance of 100 mH, a capacitance of 100 μF and a resistance of 120 Ω as shown in figure. The time in which the resistance having a thermal capacity 2 J/°C will get heated by 16°C is _____ s.

Open solution



Q64. Numerical

The position vector of 1 kg object is $\vec{r} = \left(3\hat{i} - \hat{j}\right)m$ and its velocity $\vec{v} = \left(3\hat{j} + \hat{k}\right)ms^{-1}$. The magnitude of its angular momentum is $\sqrt{x}$ Nm where x is _____.

Open solution

Q65. Numerical

A man of 60 kg is running on the road and suddenly jumps into a stationary trolly car of mass 120 kg. Then, the trolly car starts moving with velocity 2 ms^{-1}. The velocity of the running man was _____ ms^{-1}, when he jumps into the car.

Open solution

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Q66. Numerical

A hanging mass M is connected to a four times bigger mass by using a string-pulley arrangement, as shown in the figure. The bigger mass is placed on a horizontal ice-slab and being pulled by 2 Mg force. In this situation, tension in the string is $\frac{x}{5}$ Mg for x = _____. Neglect mass of the string and friction of the block (bigger mass) with ice slab.

(Given g = acceleration due to gravity)



Open solution

Q67. Numerical

The total internal energy of two mole monoatomic ideal gas at temperature T = 300 K will be _____ J. (Given R = 8.31 J/mol.K)

Open solution

Q68. Numerical

A singly ionized magnesium atom (A = 24) ion is accelerated to kinetic energy 5 keV, and is projected perpendicularly into a magnetic field B of the magnitude 0.5 T. The radius of path formed will be _____ cm.

Open solution

Q69. Numerical

A telegraph line of length 100 km has a capacity of 0.01 μF/km and it carries an alternating current at 0.5 kilo cycle per second. If minimum impedance is required, then the value of the inductance that needs to be introduced in series is _____ mH. (if $\pi = \sqrt{10}$)

Open solution

Out of syllabus 21 questions

These questions are on topics no longer in the current syllabus. They are kept for practice, after everything else.

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Q70. MCQ (Single Correct Answer) · **Chemistry**

The incorrect statement about the imperfections in solids is :

(A) Schottky defect decreases the density of the substance.

(B) Interstitial defect increases the density of the substance.

(C) Frenkel defect does not alter the density of the substance.

(D) Vacancy defect increases the density of the substance.

Open solution

Q71. MCQ (Single Correct Answer) · **Chemistry**

The Zeta potential is related to which property of colloids?

(A) Colour

(B) Tyndall effect

(C) Charge on the surface of colloidal particles

(D) Brownian movement

Open solution

Q72. MCQ (Single Correct Answer) · **Chemistry**

Given are two statements one is labelled as Assertion A and other is labelled as Reason R.

Assertion A : Magnesium can reduce Al_2O_3 at a temperature below 1350°C, while above 1350°C aluminium can reduce MgO.

Reason R : The melting and boiling points of magnesium are lower than those of aluminium.

In light of the above statements, choose most appropriate answer from the options given below :

(A) Both A and R are correct, and R is correct explanation of A.

(B) Both A and R are correct, but R is NOT the correct explanation of A.

(C) A is correct R is not correct.

(D) A is not correct, R is correct.

Open solution

Q73. MCQ (Single Correct Answer) · **Chemistry**

Dihydrogen reacts with CuO to give :

(A) CuH_2 (B) Cu (C) Cu_2O (D) $Cu(OH)_2$

Open solution

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Q74. MCQ (Single Correct Answer) · **Chemistry**

Open solution

Given below are two statements :

Statement I : The pentavalent oxide of group-15 element, E_2O_5, is less acidic than trivalent oxide, E_2O_3, of the same element.

Statement II : The acidic character of trivalent oxide of group 15 elements, E_2O_3, decreases down the group.

In light of the above statements, choose most appropriate answer from the options given below :

(A) Both Statement I and Statement II are true.

(B) Both Statement I and Statement II are false.

(C) Statement I is true, but Statement II is false.

(D) Statement I is false, but Statement II is true.

Q75. MCQ (Single Correct Answer) · **Chemistry**

Open solution

Which amongst the following is not a pesticide?

(A) DDT

(B) Organophosphates

(C) Dieldrin

(D) Sodium arsenite

Q76. MCQ (Single Correct Answer) · **Chemistry**

Open solution

Which one of the following is NOT a copolymer?

(A) Buna-S

(B) Neoprene

(C) PHBV

(D) Butadiene-styrene

Q77. MCQ (Single Correct Answer) · **Mathematics**

Open solution

If

$$\sum_{k=1}^{31}\left({}^{31}C_k\right)\left({}^{31}C_{k-1}\right)-\sum_{k=1}^{30}\left({}^{30}C_k\right)\left({}^{30}C_{k-1}\right)=\frac{\alpha(60!)}{(30!)(31!)},$$

where $\alpha \in$ R, then the value of 16α is equal to

(A) 1411

(B) 1320

(C) 1615

(D) 1855

Q78. MCQ (Single Correct Answer) · **Mathematics**

Open solution

Let A be a matrix of order 3×3 and det (A) = 2. Then det (det (A) adj (5 adj (A^3))) is equal to _____.

(A) 512×10^6

(B) 256×10^6

(C) 1024×10^6

(D) 256×10^{11}

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Q79. MCQ (Single Correct Answer) · **Mathematics**

Let the eccentricity of the hyperbola $H : \frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$ be $\sqrt{\frac{5}{2}}$ and length of its latus rectum be $6\sqrt{2}$. If $y = 2x + c$ is a tangent to the hyperbola H, then the value of c^2 is equal to :

(A) 18 (B) 20 (C) 24 (D) 32

Open solution

Q80. MCQ (Single Correct Answer) · **Mathematics**

If the tangents drawn at the points $O(0, 0)$ and $P\left(1 + \sqrt{5}, 2\right)$ on the circle $x^2 + y^2 - 2x - 4y = 0$ intersect at the point Q, then the area of the triangle OPQ is equal to :

(A) $\frac{3+\sqrt{5}}{2}$ (B) $\frac{4+2\sqrt{5}}{2}$ (C) $\frac{5+3\sqrt{5}}{2}$ (D) $\frac{7+3\sqrt{5}}{2}$

Open solution

Q81. MCQ (Single Correct Answer) · **Mathematics**

If two distinct point Q, R lie on the line of intersection of the planes $-x + 2y - z = 0$ and $3x - 5y + 2z = 0$ and $PQ = PR = \sqrt{18}$ where the point P is (1, −2, 3), then the area of the triangle PQR is equal to :

(A) $\frac{2}{3}\sqrt{38}$ (B) $\frac{4}{3}\sqrt{38}$ (C) $\frac{8}{3}\sqrt{38}$ (D) $\sqrt{\frac{152}{3}}$

Open solution

Q82. MCQ (Single Correct Answer) · **Mathematics**

The acute angle between the planes P_1 and P_2, when P_1 and P_2 are the planes passing through the intersection of the planes $5x + 8y + 13z - 29 = 0$ and $8x - 7y + z - 20 = 0$ and the points (2, 1, 3) and (0, 1, 2), respectively, is :

(A) $\frac{\pi}{3}$ (B) $\frac{\pi}{4}$ (C) $\frac{\pi}{6}$ (D) $\frac{\pi}{12}$

Open solution

Q83. MCQ (Single Correct Answer) · **Mathematics**

Let the plane $P : \vec{r}.\vec{a} = d$ contain the line of intersection of two planes $\vec{r}.\left(\hat{i} + 3\hat{j} - \hat{k}\right) = 6$ and $\vec{r}.\left(-6\hat{i} + 5\hat{j} - \hat{k}\right) = 7$. If the plane P passes through the point $\left(2, 3, \frac{1}{2}\right)$, then the value of $\frac{|13\vec{a}|^2}{d^2}$ is equal to :

(A) 90 (B) 93 (C) 95 (D) 97

Open solution

Q84. MCQ (Single Correct Answer) · **Mathematics**

Let AB and PQ be two vertical poles, 160 m apart from each other. Let C be the middle point of B and Q, which are feet of these two poles. Let $\frac{\pi}{8}$ and θ be the angles of elevation from C to P and A, respectively. If the height of pole PQ is twice the height of pole AB, then $\tan^2\theta$ is equal to

(A) $\frac{3-2\sqrt{2}}{2}$ (B) $\frac{3+\sqrt{2}}{2}$ (C) $\frac{3-2\sqrt{2}}{4}$ (D) $\frac{3-\sqrt{2}}{4}$

Open solution

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Q85. MCQ (Single Correct Answer) · **Mathematics**

Let p, q, r be three logical statements. Consider the compound statements

$S_1 : ((\sim p) \vee q) \vee ((\sim p) \vee r)$ and

$S_2 : p \rightarrow (q \vee r)$

Then, which of the following is NOT true?

(A) If S_2 is True, then S_1 is True

(B) If S_2 is False, then S_1 is False

(C) If S_2 is False, then S_1 is True

(D) If S_1 is False, then S_2 is False

Open solution

Q86. Numerical · **Mathematics**

Let l be a line which is normal to the curve $y = 2x^2 + x + 2$ at a point P on the curve. If the point Q(6, 4) lies on the line l and O is origin, then the area of the triangle OPQ is equal to _____.

Open solution

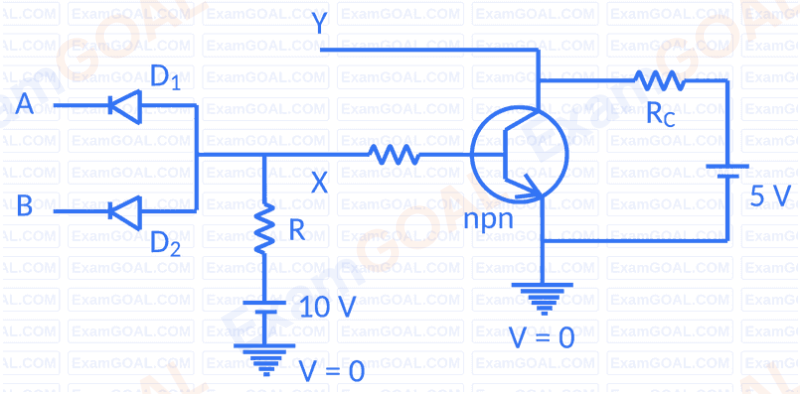
Q87. Numerical · **Mathematics**

Let the lines $y + 2x = \sqrt{11} + 7\sqrt{7}$ and $2y + x = 2\sqrt{11} + 6\sqrt{7}$ be normal to a circle $C : (x - h)^2 + (y - k)^2 = r^2$. If the line $\sqrt{11}y - 3x = \frac{5\sqrt{77}}{3} + 11$ is tangent to the circle C, then the value of $(5h - 8k)^2 + 5r^2$ is equal to _____.

Open solution

Q88. MCQ (Single Correct Answer) · **Physics**

In the following circuit, the correct relation between output (Y) and inputs A and B will be :



(A) Y = AB

(B) Y = A + B

(C) $Y = \overline{AB}$

(D) $Y = \overline{A + B}$

Open solution

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Q89. MCQ (Single Correct Answer) · **Physics**

Open solution

Given below are two statements : One is labelled as Assertion A and the other is labelled as Reason R.

Assertion A : n-p-n transistor permits more current than a p-n-p transistor.

Reason R : Electrons have greater mobility as a charge carrier.

Choose the correct answer from the options given below :

(A) Both A and R are true, and R is correct explanation of A.

(B) Both A and R are true but R is NOT the correct explanation of A.

(C) A is true but R is false.

(D) A is false but R is true.

Q90. MCQ (Single Correct Answer) · **Physics**

Open solution

Match List-I with List-II

	List-I		List-II
(A)	Television signal	(I)	03 KHz
(B)	Radio signal	(II)	20 KHz
(C)	High Quality Music	(III)	02 MHz
(D)	Human speech	(IV)	06 MHz

(A) A-I, B-II, C-III, D-IV

(B) A-IV, B-III, C-I, D-II

(C) A-IV, B-III, C-II, D-I

(D) A-I, B-II, C-IV, D-III

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Answer Key

Chemistry

1. **A**	2. **B**	3. **C**	4. **B**	5. **C**
6. **B**	7. **C**	8. **A**	9. **C**	10. **C**
11. **B**	12. **B**	13. **D**	14. **13**	15. **300**
16. **1**	17. **0**	18. **14**	19. **3**	20. **6**
21. **16**	22. **3**	23. **34**		

Mathematics

24. **D**	25. **B**	26. **D**	27. **C**	28. **C**
29. **C**	30. **B**	31. **A**	32. **A**	33. **B**
34. **A**	35. **8**	36. **2**	37. **17**	38. **150**
39. **31**	40. **702**	41. **2**	42. **40**	

Physics

43. **C**	44. **C**	45. **A**	46. **A**	47. **C**
48. **A**	49. **A**	50. **A**	51. **A**	52. **D**
53. **B**	54. **A**	55. **C**	56. **A**	57. **D**
58. **B**	59. **D**	60. **5**	61. **19**	62. **10**
63. **15**	64. **91**	65. **6**	66. **6**	67. **7479**
68. **10**	69. **100**			

Out of syllabus

70. **D**	71. **C**	72. **B**	73. **B**	74. **D**
75. **D**	76. **B**	77. **A**	78. **A**	79. **B**
80. **C**	81. **B**	82. **A**	83. **B**	84. **C**
85. **C**	86. **13**	87. **816**	88. **C**	89. **A**
90. **C**				

For numerical answers given as a range, any value within the range is correct. Full solutions: scan the QR code next to each question, or visit questions.examside.com.

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